
CAPITAL STRUCTURE

DECISION MAKING: A

MODEL FOR FAMILY

BUSINESS

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EXECUTIVE SUMMARY

Most theoretical and empirical studies of capital structure focus on public corporations. Only a limited number of studies on capital structure have been conducted on small-to-medium size enterprises (SMEs), and this deficiency is particularly evident in investigations into factors that influence funding decisions of family business owners.

Theory indicates that there is a complex array of factors that influence SME owner-managers' financing decisions. Recent family business literature suggests that these processes are influenced by firm owners' attitudes toward the utility of debt as a form of funding as moderated by external environmental conditions (e.g., financial and market considerations).

A number of other factors have been shown to influence financing decisions including culture; entrepreneurial characteristics; entrepreneurs' prior experiences in capital structure; business goals; business life-cycle issues; preferred ownership structures; views regarding control, debt-equity ratios, and short- vs. long-term debt; age and size of the firm; sources of funding for growth; attitudes

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toward debt financing; issues relating to independence and control; and perceived risk and attitudes toward personal risk.

Although these factors have been identified, until now there does not appear to have been any attempts to develop empirically-based models that show relationships between these factors and family business owners' financing decisions. Utilizing theories derived from divergent disciplines, this study develops an empirically tested structural equation model of financing antecedents of family businesses. Participants of our study involved a random sample of 5000 business owners who were mailed a 250-item Australian Family and Private Business questionnaire developed specifically for this investigation.

Notably, our findings reveal that firm size, family control, business planning, and business objectives are significantly associated with debt. Small family businesses and owners who do not have formal planning processes in place tend to rely on family loans as a source of finance. However, family businesses in the service industry (e.g., retailers and wholesalers) are less likely to use family loans as are those owners who are planning to achieve growth through new products or process development. Use of capital and retained profits is likely for family businesses planning to achieve growth through an increase in sales but less is likely for family businesses in the manufacturing sector and lifestyle firms. In addition, debt and family loans are negatively related to capital and retained profits. Equity is a consideration for owners of large businesses, young firms, and owners who plan to achieve growth through increasing profit margins. However, equity is less likely to be a consideration for older family business owners and owners who have a preference for retaining family control.

Our findings suggest that the interplay between multiple social, family, and financial factors is complex. In addition, our findings indicate the importance of utilizing theories that also help to explain behavioral factors (e.g., owners' needs to be in control) that affect financial structure decision-making processes. Practitioners and researchers should consider the dynamic interplay among business characteristics (e.g., size or industry), behavioral aspects of business financing (e.g., business objectives), and financial factors (e.g., gearing levels) when working with and researching family enterprises. © 2000 Elsevier Science Inc.

INTRODUCTION

Family businesses¹ in Australia and elsewhere have a major impact on growth of economies through generation of employment, productivity, and innovation (Francis 1993; Smyrniotis and Romano 1994; Smyrniotis, Romano, and Tanewski 1997). Although there has been considerable interest in family businesses, research appears to have lacked empirical rigor (Westhead and Cowling 1996). Moreover, business and organizational literature in this area has not only been limited but is predominantly anecdotal and descriptive (Brockhaus 1994). This deficiency is particularly evident in investigations into factors that influence funding decisions of family business owners. Recent exceptions include Gallo and Vilaseca (1996) and Poutziouris, Chittenden, and Michaelas (1998). There is also little empirical evidence that identifies specific variables affecting business owners' financing decisions (Romano and Ratnatunga 1994). In contrast to the family business domain, there is considerable theoretical and empirical work in finance and

¹In view of the pertinent literature, Wortman (1995) reported that there are over 20 definitions of family business. Given a lack of definitional clarity (Wortman 1994), and in line with research (Smyrniotis and Romano, 1994; Smyrniotis et al., 1997; Stoy Hayward and the London Business School 1989, 1990), we have broadly defined a family firm as one in which any of the following three criteria holds true: 50% or more of the ownership is held by a single family or by multiple members of a number of families; a single family group is effectively controlling the business; and a significant proportion of senior management is drawn from the same family.

economic disciplines relating to capital structure decision-making processes (e.g., Chaganti, DeCarolis, and Deeds 1995; Harris and Raviv 1991; Stulz 1990).

A review of the extant interdisciplinary literature indicates that there is a complex array of factors that influence small-to-medium size enterprise (SME) owner-managers' financing decisions. However, it appears that this literature has not made explicit those factors that influence family business owners' decisions on the choice of different forms of finance, such as debt or equity. This failure can be attributed to most research (e.g., Harris and Raviv 1991) focusing primarily on either public companies or privately-held SMEs with scant consideration of family issues.

Another reason relates to the emphasis on single equation or recursive systems (e.g., Harris and Raviv 1991). Yet, a further reason involves the application of divergent theoretical frameworks to drive research with little, if any, attempt to integrate these theories. These frameworks include capital structure theories grounded in finance paradigms such as agency and transaction cost theories (e.g., Jensen and Meckling 1976; Miller 1977; Modigliani and Miller 1958); managerial perspectives that focus on nonfinancial and behavioral factors (e.g., Barton and Matthews 1989); strategic and corporate perspectives that take into account managerial characteristics (e.g., Barton and Gordon 1987, 1988); and the theory of reasoned action (e.g., Ajzen and Fishbein 1980; Fishbein and Ajzen 1975) whereby personal values and behavioral intentions are viewed as determining behavior. This theoretical divergence reflects Myers (1984) and Norton's (1991) view that neither financial theories nor empirical studies incontrovertibly explain critical factors that affect owners' or managers' financing decisions.

Given these issues, this investigation adopts an interdisciplinary framework and aims to integrate divergent theories within a broad-based framework to help explain financing decisions from a family business perspective. The present study identifies antecedent organization and owner-manager characteristics in order to develop a model that identifies factors that influence owner-managers' choice of either debt, equity, family loans, or securities as different forms of finance.

BACKGROUND AND THEORY

Importance of the Present Research and Rationale

Over 15 years ago, Myers (1984) stated that financial theories do not adequately explain financing behavior. Despite this acknowledgment, the key determinants of capital structures of SMEs do not appear much clearer today (Chaganti, DeCarolis, and Deeds 1995). Obviously, without a clear appreciation of the magnitude and direction of relationships between antecedent organization and owner-manager characteristics and capital structure, researchers are unable to explicate satisfactorily how family business owners choose between different types of finance. For example, based on anecdotal evidence, Ward (1987, p. 3) maintained that family business owners typically reinvest most, if not all, of their funds during the early stages of the life cycle of their business. However, in later years, because of families' growing financial demands, owners tend to use company profits rather than reinvesting capital for additional growth. Does this view mean that in the early stages of a family enterprise a business is financed predominantly by owners' savings and that in later years owners pursue alternative sources of finance? This point leads to another question. What determines a family firm's capital structure?

The latter question suggests that explanations might require an understanding of organizational structure and environment, such as levels of ownership and management control (Boyer and Roth 1978; Ray and Hutchinson 1983), years of business establishment (Stanworth and Curran 1976), and business owners' plans and objectives (McMahon and Stanger 1995). These considerations echo Donaldson and Lorsch (1983) who highlighted a dynamic interaction of elements within and external to firms, further underscoring a need for researchers to focus on the multifaceted nature of family businesses, including factors that have an impact on firms such as owners' business, social, and behavioral goals (McMahon and Stanger 1995); family values and aspirations (e.g., succession); corporate and business planning (Storey 1994); and industry considerations (Carleton and Silberman 1977).

Barton (1989) also identified financial, personal, and social variables that influence family business owners' capital structure decisions. Factors singled out included entrepreneurs' prior experiences in capital structure; preferred ownership structures (e.g., employee stock options); use of internal financing (e.g., to clear debt); views regarding control, debt–equity ratios, and short- vs. long-term debt; age of the firm; perceived key sources of funding for growth (e.g., retained earnings vs. debt); attitudes toward debt financing; and perceived risk. Thus, if organization and owner-manager background characteristics are found to influence capital structure decision making, then this study will provide a basis for explaining financing behavior of family firms and contribute substantially to the emerging academic field of financial management.

Theoretical Rationale

Contemporary theory of capital structure is based on Modigliani and Miller's (1958) seminal work into the effect of capital structure on company value. This theory assumes perfect markets and perfect competition in which companies operate without taxes or transaction costs and where all relevant information is available without cost. Under these conditions, Modigliani and Miller (1958) argued that modifying a company's capital structure does not change the companies' value or shareholders' wealth. Changing a companies' capital structure simply alters ways in which streams of net operating cash flow is divided between different classes of investors.

In relation to debt–equity ratios of firms, Modigliani and Miller (1958) proposed that financial leverage is not related to shareholder wealth, that shareholders' expected rates of return are directly related to levels of gearing (i.e., debt–equity ratio), and that expected increases in returns are negatively related to risk and shareholders' required ratios of return.

Hence, in a perfect capital market, only investment decisions are important in pursuit of wealth maximization. However, when these assumptions are relaxed, factors that could make capital structure important include taxes, agency costs, costs of financial distress, and information asymmetry. Notwithstanding, Modigliani and Miller have been criticized on the grounds that their theory assumes rational economic behavior and perfect market conditions, owners' goals are targeted only at maximizing profits (Grabowski and Mueller 1972), and that it has limited applicability to small and family firms (Chaganti, DeCarolis, and Deeds 1995).

Following Jensen and Meckling's (1976) work into the theory of the firm, agency theory provides valuable insights into small- and family-firm finance. Jensen and Meckling (1976) argued that operations of companies involve sets of contractual or agency

relationships between various parties such as owners, managers, creditors, customers, and suppliers. These relationships involve agency costs, which arise from the potential for conflicts of interest between parties, such as owners and managers, owners and creditors, as well as incentive effects of debt. In addition, Jensen and Meckling argued that an optimal capital structure could be obtained by trading off agency costs of debt against any benefits of debt. In short, agency theory predicts that firms in mature industries with few growth opportunities (e.g., airlines and steel) are highly leveraged, whereas firms with large cash inflows and which experience slow or negative growth (e.g., mining) have more debt.

Such agency cost arguments have led to propositions suggesting that a trade-off between the opposing effects of benefits of debt finance and the costs of financial distress means that an optimal capital structure exists. In contrast, *trade-off* theory (e.g., Harris and Raviv 1990; Jensen and Meckling 1976; Stulz 1990), which is based on the effects of taxes, agency costs, and costs of financial distress, suggests that management aims to maintain a target debt–equity ratio. In other words, a firm’s optimal debt ratio is determined by a trade-off of costs and benefits of borrowing.

An alternative approach proposes that, in raising finance, managers follow a pecking order in which internal funds are preferred, followed by debt, hybrid securities, and then, as a last resort, a new issue of ordinary shares. The *Pecking Order* approach (cf. Donaldson 1961; Myers 1984) does not rely on the existence of a target debt–equity ratio. Rather, a company’s actual debt–equity ratio varies over time, depending on its need for external finance. This approach can be explained by a desire to minimize the transaction costs of raising finance.

In many instances, owner-managers make decisions that take into account a conglomeration of competing management, family, economic, market, and industry considerations. The capital needs of family businesses can be satisfied in a number of ways: through internally generated cash flows; additional capital injections by current shareholders; by broadening the circle of shareholders (either through a share float or by inviting employees, directors, or investment institutions to buy shares); through loans from insiders and/or third parties; and by selling parts of the business that do not belong to the core activities of a firm (Neubauer and Lank 1998). This sequencing of funding in a family business can be viewed in the context of a financial pecking order.

Myers (1984) provides yet another explanation based on *information asymmetry* in which firm managers or insiders are assumed to possess private information about the characteristics of a firm’s return stream or investment opportunities. Myers and Majluf (1984) showed that if investors are less well informed than current firm insiders about the value of a firm’s assets, then equity can be mispriced by the market. Thus, as greater risk is attached to a firm, leverage increases in accord with the extent of information asymmetry.

In summary, most of these theories propose that there is an optimal capital structure that maximizes the value of a company. However, propositions relating to profit maximization and assumptions of perfect markets and rational economic behavior have limited the validity of these theories, casting some doubt on their ability to explain capital structure decisions in both large and small enterprises (Barton and Gordon 1987; Myers 1984). Indeed, Myers (1984) argued that “we know very little about capital structure . . . [and] our theories don’t seem to explain actual financing behavior” (p. 575). Clearly, any paradigm that attempts to explain capital structure decision-making processes must go beyond the use of publicly available information found on extensive data-

bases or in annual reports to include matters that most owner-managers might regard as privileged information (McMahon and Stanger 1995; Neubauer and Lank 1998).

Existing capital structure theories do not include factors such as owners' objectives and business planning decisions, which are relevant to financing decisions. Because family businesses do not have to respond to market scrutiny, preferences and objectives of owner-managers in these firms take on significance in capital structure decisions (Barton and Matthews 1989; McMahon and Stanger 1995). Matthews, Vasudevan, Barton, and Apana (1994) and Barton and Matthews (1989) proposed that a firm's capital structure is affected by managements' goals (see also Chaganti, DeCarolis, and Deeds 1995). These investigators suggest that the amount of debt sought is determined, to a large extent, by entrepreneurs' goals of protecting a proprietary process. Similarly, Hutchinson (1995) argued that factors limiting demand for finance in owner-managed firms "have often been neglected by an over-reliance on arguments which attempt to explain small firm financing problems solely in terms of a lack of access to the supply of equity and debt" (p. 238). Accordingly, explanations of capital structure processes should be sought by understanding the impact of owner-managers' personal preferences and values and of the firm characteristics.

Of late, two theoretical positions (Dreux 1990; Matthews et al. 1994) have emerged on issues concerned specifically with financing family businesses. On the one hand, Dreux (1990) developed a general conceptual framework that provides a perspective for simplifying the conflicting objectives of businesses and shareholders. This conception identifies a range of competing scenarios (e.g., the growth of a business vs. the retirement of an owner) and competitive interactions between fundamental interests of the family business shareholders and the business (e.g., control, liquidity, and capacity). Dreux (1990) argued that financial objectives could not be achieved without a major recognition of the fundamental issues relating to ownership, appropriate capital levels, and the control of the business.

On the other hand, Matthews et al. (1994) proposed a framework in which capital structure decision making is influenced by owners' attitudes toward the utility of debt as a form of funding as moderated by external environmental conditions (e.g., financial and market considerations), as well as owners' needs to be in control, risk propensity, experience, social norms, and personal net worth. Drawing on these perspectives, the present study examines pertinent factors that influence owner-managers' choice of either debt, equity, family loans, or securities as alternative forms of finance. With particular emphasis on empirical research in this area, these relationships are discussed in the next section.

Empirical Evidence and Hypothesis Development

A considerable body of research has described sources of capital utilized by SME owner-managers. As noted earlier, most studies (e.g., Ferri and Jones 1979; Harvey and Evans 1995; Hutchinson 1995; Norton 1991; Van der Wijst 1989) have been either descriptive or conducted on SMEs and publicly owned firms without taking family issues into consideration. Notwithstanding, interdisciplinary and SME literature provides useful insights into sources of capital utilized by family businesses. Broadly speaking, what emerges from this review of the literature is that family business owners are averse to sources of capital that are beyond traditional commercial banking arrangements such as investment and venture capital, initial public offerings, funding through general finance

companies, and access to state and local funds (Dreux 1990; Mukherjee 1992; Poutziouris, Chittenden, and Michaelas 1998).

The following subsections explore a number of antecedent conditions, including size of the business, industry sector, firm age, family control, age of the CEO, business plans, owners' business objectives, and growth factors, in order to develop a conceptual framework within which to build and subsequently test a hypothetical model of family business capital structure decision making processes.

Size

Consistent with market efficiency hypotheses, Modigliani and Miller's (1958) theory implies that size does not affect a firm's capital structure. However, a number of studies (e.g., Berger and Udell 1998; Chittenden, Hall, and Hutchinson 1996; Hutchinson 1995; Mills and Schumann 1985) demonstrate links between size and financial structure. These investigations have used agency theory to explain unique characteristics of small firms and their effects on capital structure. Chittenden, Hall, and Hutchinson (1996) noted that the pecking order approach is particularly relevant to small businesses largely because the costs associated with external finance are higher for small businesses than they are for large firms. Chittenden, Hall, and Hutchinson (1996) also found that small and young firms rely heavily on short-term finance and that the "use of short-term debt is negatively related to asset structure" (p. 64). In line with these findings, a number of other investigators (e.g., Constand, Osteryoung, and Nast 1991; Hutchinson 1995; Mills and Schumann 1985; Renfrew, Sheehan, and Dunlop 1984) reported that, because of lower costs, small firms rely more heavily on short-term rather than long-term (e.g., equity) bank debt. Moreover, Berger and Udell (1998, p. 619) stated that commercial banks and financial companies are the primary lenders of finance to SMEs (i.e., companies employing more than 20 employees and involving sales greater than \$1 million).

Gallo and Vilaseca (1996), investigating the impact of capital structure behavior toward investment and risk, and dividend policy on performance, revealed that family firm size is associated with the use of multiple financial institutions and diverse financial products for financing and that family businesses tend to have low debt-equity levels, particularly those firms that are market leaders. According to Sonnenfeld and Spence (1989), family businesses have low debt-equity levels in order to avoid damaging their family's reputation and personal guarantees and to avoid losing everything in the case of loan failure. On the supply side, financial institutions seem to place a heavy reliance on wealth rather than on the repayment capabilities of family businesses. For example, Berger and Udell (1998) argued that banks or commercial finance companies place a heavy weight on production levels and substantial business assets that might be pledged as collateral (see also Mukherjee 1992).

In a descriptive study of long-term financing arrangements of Australian small businesses, Renfrew, Sheehan, and Dunlop (1984) found that banks are the main source of finance for small firms, with more than four-fifths of owners having access to bank overdrafts. Interestingly, small and young firms are less likely to apply for financing. Similar financing patterns were disclosed by an Industry Commission (1991) report on the availability of capital to Australian business. This report noted that banks are the major source of debt finance for small businesses, followed by finance companies and trade credit. The commission also observed that small businesses pay significantly more for debt finance than large businesses. These findings are consistent with those reported

in the United States (Mukherjee 1992). Accordingly, the following hypotheses are proposed.

H1A: The size of family firm is associated positively with debt.

H1B: The size of family firm is associated positively with capital and retained profits.

H1C: The size of family firm is associated positively with equity.

Industry

Harris and Raviv (1991) noted that firms within a particular industry are more similar in their capital structure than those in different sectors. For example, Bowen, Lane, and Huber (1982) and Bradley, Jarrell, and Kim (1984) observed that, while drug, instrument, electronic, and food industry sectors have consistently low leverage, paper, textile, steel, airline, and cement industries have consistently high leverage. In a Canadian study, Riding, Haines, and Thomas (1994) revealed that manufacturing companies have significantly lower access to bank overdrafts than non-manufacturing firms. In contrast, Riding, Haines, and Thomas (1994) found that non-manufacturing firms are more likely to experience term loan application turndowns than enterprises in the manufacturing industry. Van der Wijst (1989) and Welsh and White (1981) reported that while the manufacturing industry is capital intensive and requires large investments in fixed assets derived from both debt and equity, negative relationships between short-term debt and the retail sector have been noted. Therefore, the following related hypotheses are proposed.

H2A: Family firms in the manufacturing industry are associated positively with debt.

H2B: Family firms in the manufacturing industry are associated positively with capital and retained profits.

H2C: Family firms in the manufacturing industry are associated positively with equity.

H2D: Family firms in the retail industry are associated negatively with debt.

Age of the Firm

Different financing arrangements have also been linked with business life-cycle issues (e.g., Berger and Udell 1998; Gersick, Davis, McCollom Hampton, and Lansberg 1997). Dollinger (1995) indicated that sources of capital depend, to some extent, on whether a business is developing or maturing. For example, developing firms tend to rely on equity because of difficulties raising debt, whereas mature businesses are able to leverage assets to raise debt. In support of these views, Kimki (1997) argued that self-financing and the ability of a family to acquire debt was limited during the early stages of a family business. Thus, it is not uncommon for entrepreneurs to rely on personal savings, friends, and relatives as primary sources of capital at startup (Hutchinson 1995; Wal-dinger, Aldrich, and Ward 1990).

Berger and Udell (1998) suggested that firms can be viewed as lying on a size–age continuum, arguing that small and young firm owner-managers rely on internal finance such as family loans, trade credit, and/or business angel finance. Collins and Moore (1964) reported that first-generation owners were more likely to be opposed to external borrowings. However, these findings should be tempered to some extent by research

findings (e.g., Stanworth and Curran 1976), which indicate that fast-growth firms rely on this type of finance. Similarly, Berger and Udell (1998) found that small businesses in high-growth, high-risk sectors relied on equity, whereas small businesses in low-growth sectors relied on debt finance.

Bates (1991) found that when compared with older more established businesses, younger firms rely less on profits derived from sales. However, as firms grow and mature, different types of debt arrangements (first short-term, then long-term) become important until a firm is ready to enter the public issue market and obtain outside equity (Berger and Udell 1998; van der Wijst 1989). These findings are reflected in a relatively recent National Investment Council (1995) report to the Australian Federal Government on financing growth. This report explained that at inception businesses' major sources of finance are owner funds, with these being supplemented by relatives and friends. At growth and maturity stages, bank funding, secured against personal assets, tends to be employed, while equity is used for the expansion and growth of an enterprise. Thus, the following two hypotheses are proposed.

H3A: The age of a family firm is associated positively with debt.

H3B: The age of a family firm is associated negatively with family loans.

Family Control

Neubauer and Lank (1998) maintained that business ownership, independence, and family control factors affect owners' financing decisions (see also Dreux 1990). The literature (e.g., Dailey, Ruschling, and De Mong 1977; Hutchinson 1995; Shrivastava and Grant 1985) suggests that entrepreneurs who have a strong preference for independence tend to utilize equity (as long as it does not involve outside participation) or retained earnings as sources of finance. Similarly, Storey (1994) stated that small business owners tend to use short-term debt financing because they are strongly opposed to sharing ownership. According to Kotkin (1984), use of venture capital as a source of capital raises fears of loss of control. Hutchinson (1995, p. 238) also pointed out that owner-managers who have a strong desire to retain control of the firm "may actively place limits on the use and growth of equity." This view is supported by Berger and Udell (1998), Harvey and Evans (1995), and Barton (1989), who found that issues relating to control and to risk aversion influence capital structure and financing decisions. Modigliani and Miller (1958) argued that creditors restrict top management's flexibility and prerogatives by stipulating rigid terms regarding capital structure. Such covenants have culminated in entrepreneurs being apprehensive about loss of control (Shrivastava and Grant 1985). Notwithstanding this, a high percentage of business failures can be attributed to unsound financing decisions including undercapitalization and inappropriate capital structures, such as an inappropriate mix of debt vs. equity finance (Jones 1979).

H4A: Owners who indicate a preference for retaining family control are associated positively with family loans.

H4B: Owners who indicate a preference for retaining family control are associated positively with capital and retained profits.

H4C: Owners who indicate a preference for retaining family control are associated negatively with equity.

Age of the CEO

In terms of entrepreneur characteristics, small business finance literature suggests that older entrepreneurs tend to be less willing to invest additional finances into their firms. Van der Wijst (1989) found that older entrepreneurs use less debt and are more reluctant than younger entrepreneurs to accept outside participation. Exceptions are older entrepreneurs who lack familial successors and are prepared to accept nonfamily participants as possible successors (Ward 1987).

H5A: The age of the owner (CEO) is related negatively to debt.

H5B: The age of the owner (CEO) is related negatively to capital and retained profits.

H5C: The age of the owner (CEO) is related negatively to equity.

Business Planning

It is not uncommon for small business owners to use, at startup, formal business plans as a sales document to obtain debt or other external forms of finance (Berger and Udell 1998). In line with this proposition, Harvey and Evans (1995) found that when evaluating loans, banks emphasize the importance of effective business planning, which has been shown to be positively related to debt.

H6: Family firms that report having written business plans in place are associated positively with debt.

Business Objectives and Plans to Achieve Growth

Owner-managers of firms tend not to have a single overriding objective when establishing or running a business (McMahon and Stanger 1995). Business objectives, like personal values, are complex and extend beyond pecuniary interests. Owners' business objectives have been shown to be associated with growth orientation, firm size, legal structure(s), sectoral consideration, stage of business development, and entrepreneurial characteristics (Chaganti, DeCarolis, and Deeds 1995).

Consistent with these findings, Boyer and Roth (1978) reported that, with the exception of growth or entrepreneurial firms in which liquidity and maximizing profits and sales growth are of primary concern, many owner-managers emphasize nonpecuniary concerns such as control, lifestyle, and job security over rates of return on their investments (see also Petty and Bygrave 1993; Ray and Hutchinson 1983; Stanworth and Curran 1976). Not surprisingly, Barton and Gordon (1987) argued that financing decisions are determined to a large extent by owners' values, business objectives, and aspirations.

For many family business owners, business growth is not an objective (Curran 1986). These firms are typically home-based and employ only a few people (Hakim 1989). In contrast, growth firms, and in particular fast growing businesses, not only actively look for significant expansion, but also have been identified as likely purchasers of a wide range of financial services including loans and equity capital (Storey 1994). Research demonstrates a positive association between growth and external equity. For example, Chaganti, DeCarolis, and Deeds (1995) noted that entrepreneurs who are "bullish" about their business tend to seek equity rather than debt financing. However, van der Wijst (1989) stated that entrepreneurs who aim principally to grow their busi-

ness tend to use a comparatively large portion of debt-to-finance business development. By comparison, entrepreneurs whose objective is to use the business for "steady employment" rely on debt finance.

Van Auken and Carter (1989) suggested that owner-managers prefer to use internal sources of capital rather than equity or debt finance when their businesses are profitable. Myers (1984) argued that profitable firms prefer to fund their operations with internal equity. In line with this view, van der Wijst (1989) found that profitability and retained earnings are negatively related to debt. In comparison, Constand, Osteryoung, and Nast (1991) found a positive relationship between profitability and debt, and this source of finance is relevant for both small and large firms. Meanwhile, Ang (1992) pointed out that an important objective of profitable small enterprises, particularly those that do not use external finance, is to maximize the long-term value of their business.

H7A: Family firms whose objective is to increase the value of their business are associated positively with debt.

H7B: Family firms whose objective is to increase the value of their business are associated positively with capital and retained profits.

H7C: Family firms whose objective is to increase the value of their business are associated positively with equity.

H8A: Family firms whose objective is to grow their business are associated positively with debt.

H8B: Family firms whose objective is to grow their business are associated positively with capital and retained profits.

H8C: Family firms whose objective is to grow their business are associated positively with equity.

H9: Debt and family loans are negatively related to capital and retained profits.

The present study goes beyond traditional finance paradigms by incorporating elements from divergent perspectives, including family business, finance, economics, and management, to explore capital structure decision-making processes. This research led to the development of a model for understanding ways in which characteristics relating to family business owners (e.g., attitudes toward stock exchange listing), family (e.g., family control), and business (e.g., firm age and industry type) interact to affect financing decisions. To date, it appears that such a paradigm has not been employed. Figure 1 shows hypothesized directions of relationships among exogenous and endogenous variables. Appendix 1 includes definitions of variables used in this investigation.

METHOD

Data Collection Procedure

A random sample of 5000 businesses based on state of location, industry, and sales turnover was obtained from Dun and Bradstreet (1996). Questionnaires were mailed in December 1996 and January 1997 to business owners with an accompanying letter explaining the purpose of this study. Questionnaires were returned in stamped, self-addressed envelopes. The valid response rate to this survey was 29.2% ($n = 1490$) and

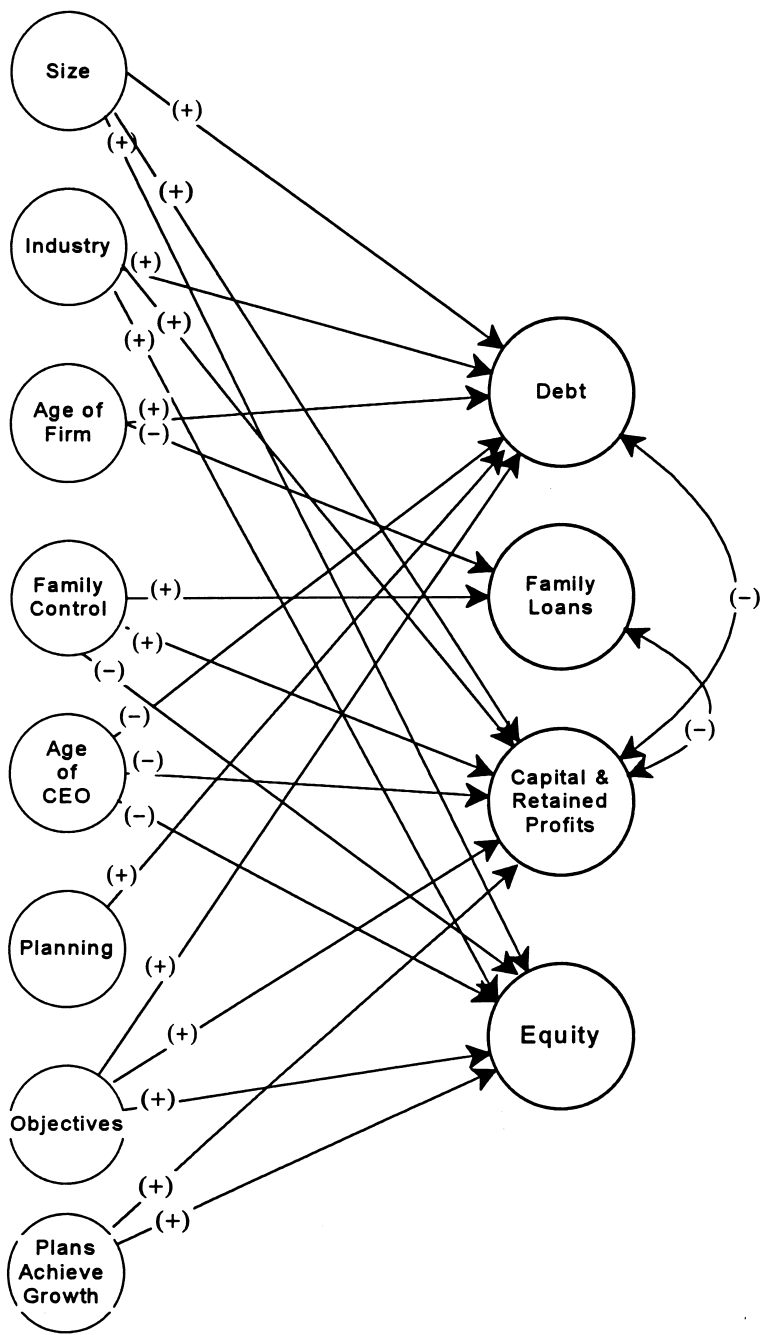


FIGURE 1 Hypothesized model for family business finance antecedents and outcomes.

includes nonuseable return questionnaires ($n = 34$), businesses that indicated they were not family firms ($n = 368$), and *return-to-sender* mail ($n = 29$). This response rate compares favorably with those reported overseas, for example, Dunn (1995) at 15% and Nager, Aronoff, and Ward (1995) at 10%. No response bias tests were conducted to compare valid respondents with nonrespondents. This comparison was not possible because we did not have a family business sample frame prior to mail out. Notwithstanding this, characteristics of early respondents (e.g., industry, location, sales turnover, and number of employees) were not significantly different from late respondents. As noted in Statistical Procedures, correlation matrices and structural relations estimates from two randomly selected halves of this data set did not differ significantly.

Participants

Table 1 shows information on background variables including age of owners, country of origin, education level, marital status, industry type, legal structure of the businesses, length of business establishment, generation of family ownership, number of employees, gross sales, and average rates of growth in sales over the previous 3 years. Interestingly, proportions relating to characteristics of family business owners are in line with those reported by the Australian Bureau of Statistics (1997), Smyrnios and Romano (1994), and studies from overseas (e.g., Dunn 1995; Nager, Aronoff, and Ward 1995; Stoy Hayward and The London Business School 1990).

Australian Family and Private Business Questionnaire

For the purposes of the present study, a 250-item questionnaire was developed by a team of academic researchers, family business consultants, and family business owners. The Australian Family and Private Business Questionnaire (AFPBQ) comprises ten sections: background of business, current ownership, management of business, succession and retirement plans, family business issues, banking and insurance, business growth planning, management development and training, alternative investment, and background of the owner (see Smyrnios, Romano, and Tanewski 1997 for a detailed description of the AFPBQ). This questionnaire was adapted from three principal sources: Dunn (1995), Smyrnios and Romano (1994), and Stoy Hayward and The London Business School (1989, 1990). The AFPBQ was pilot tested on 10 family businesses and among academic colleagues to determine its face validity and to identify any issues of concern. Scales of the AFPBQ demonstrate high levels of internal reliability (Smyrnios, Tanewski, and Romano 1998).

Statistical Procedures

Data analyses proceeded through four principal stages: data screening, a principal components analysis (PCA) with orthogonal rotation, a confirmatory factor analysis (CFA) of factors derived from the PCA, and Structural Equation Modelling (SEM). First, data were tested for violations of statistical assumptions including sample size, normality, linearity, multicollinearity, and factorability of R . Bartlett's test of sphericity and the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy were checked ($KMO = .74$; Bartlett's test of sphericity = 1984.95, $p = .0001$). Tests resulted in nonviolations of underlying assumptions. Second, data relating to AFBQ were subjected to a PCA

TABLE 1 Characteristics of Family Business Owners

Characteristic	% (<i>n</i> = 1059)
Country of Origin of Business Owner	
Australia	73.9
Europe	16.0
Asia	1.5
Other countries (e.g., USA)	8.6
Educational Level	
Tertiary	39.9
Secondary School	30.8
Gender	
Male owner	97.2
Female owner	2.8
Marital status	
Married	87.5
Single	1.7
Divorced	5.8
Other	5.4
Age (years) of Business Owner	$\bar{X}$ = 52
<51	46.0
51–60	33.4
>60	20.8
Age (years) of Firm (Median)	
<10	22.9
11–20	28.8
21–40	27.3
>40	21.0
Number of Employees (Median)	
Full-time (Person)	34
Part-time (Person)	12
Rate of Growth (%) (Median)	15.0
Sales (\$million) (Median)	11.7
Industry	
Manufacturing	33.5
Services	44.9
Technology	7.0
Other	14.6
Ownership	
One Family	65.8
More Than One Family	15.0
Single Family Has Majority Management	13.5
Single Family Exerts Significant Control	5.7
Legal Structure	
Private Company	80.1
Family Trust	13.6
Partnership	3.4
Sole Traders	0.7
Listed on Stock Exchange	2.2
Generation of Ownership	
First Generation	70.6
Second Generation	20.3
Third Generation	6.2
Fourth Generation	2.9

with varimax rotation. PCA was conducted for three main reasons: scientific parsimony, the reduction of the number of predictors to increase the n/k ratio, and cross validation, that is, items overlapped in content (Lord and Novick 1968; Stevens 1996). Third, using the maximum-likelihood method of LISREL 7 (Jöreskog and Sörbom 1989), congeneric measurement models (CFA) were carried out to assess the unidimensionality of factors and to test whether observed correlations among AFBQ items can be explained by factors hypothesized to underlie the AFBQ. This procedure was preceded by generating through PRELIS (a data-screening program) and a LISREL preprocessor program (a correlation matrix).

Fourth, utilizing findings that demonstrate relationships among relevant variables associated with capital structure decision-making processes, we developed a hypothesized SEM that was evaluated using correlation matrices and the maximum likelihood procedure (see Figure 1). SEM, also referred to as causal modeling, examines a series of dependence relationships concurrently. SEM estimates a series of separate but interdependent multiple regression equations simultaneously. Causal relationships between two or more variables are specified and drawn from theory and research hypotheses to distinguish which independent variables predict each dependent variable.

Matrices were computed using listwise missing data. With listwise deletion, all respondents with missing data on one or more variables are deleted, ensuring that all computations are based on the same cases and that matrices are nonnegative-definite. To simplify identification of the model, variables were treated as observed rather than latent constructs. In order to control for biasing effects of measurement error, all variables in our model were assigned random error variances, θ_{δ} θ_{ϵ} , by placing random error values to the product of each measured variable (see Hayduk 1987, pp. 119–120).

Using LISREL's multisample feature, our hypothesized model was estimated simultaneously across two randomly divided halves of the data (i.e., 55% and 45%, respectively). This procedure assessed whether correlation matrices and structural relations estimates from models developed on one random half of the data set differed significantly from estimates on the other half of the data. Bollen (1989), Hayduk (1987), and Jöreskog and Sörbom (1989) suggested that this technique provides a rigorous validation of a model's structural parameters. The first multisample test assessed structural relations estimates without imposing invariance assumptions on parameters of interest. This test, which resulted in $\chi^2 = 151.37$, $df = 128$, and $p = .078$, is least demanding and merely assesses whether groups have the same form. The second test constrained elements of B and Γ to be invariant across both groups, and this test resulted in $\chi^2 = 157.80$, $df = 138$, and $p = .097$. Comparability analysis tests the null hypothesis of statistical equivalence between covariance matrices and model parameters. Both tests showed that the two randomly selected groups are not significantly different from each other, providing validity to our model's structural parameters.

RESULTS

Principal Components Analysis

To evaluate interrelationships and appropriateness of using 11 items that made up constructs of *size*, *business planning*, and *attitude toward family control*, a principal components analysis with varimax rotation was performed (see Appendix 1 for operationalization of variables). This analysis reveals three clear interpretable factors accounting for

TABLE 2 Varimax Factor Matrix Estimates of Family Business Background Items

	F1	F2	F3	h^2 *
Size of Business (F1)				
No. Employees	0.84	-0.24	0.01	0.74
Gross Sales for 1996	0.80	-0.21	-0.01	0.68
Estimated Value of Business	0.73	-0.23	-0.13	0.60
No. Locations (Total)	0.62	0.04	0.12	0.44
No. Locations (Australia)	0.62	-0.01	0.01	0.41
Business Planning (F2)				
Business Plan	-0.09	0.86	-0.03	0.74
Formal Strategic Long-Term Plan	-0.07	0.76	-0.08	0.59
Formal Management Structure	-0.29	0.69	-0.07	0.56
Owners' Attitude—Family Control (F3)				
Allow Nonfamily Shareholders	0.01	-0.05	0.87	0.76
Make Shares Available to Nonfamily	-0.01	0.08	-0.79	0.63
Benefit from External Sources of Funds	0.02	-0.03	0.67	0.45
Cronbach's Alpha (α)	0.75	0.71	0.45	
Eigenvalues	3.29	1.91	1.34	
% Total Variance	29.9	17.4	12.2	
Cumulative Variance	29.9	47.2	59.4	

* Communality estimates.

59.4% of the variance (see Table 2 for factor matrix estimates of items). Five items that reflect size (item loadings range from $r = .62$ to $r = .83$), three items that measure business planning (item loadings ranged from $r = .69$ to $r = .86$), and three items that reflect attitudes toward family control (item loadings range from $r = .67$ to $r = .87$) demonstrate adequate reliability (Cronbach's alpha [α] = .75, .71, and .45, respectively). Items measuring these three constructs were saved as factor (regression) scores and used in subsequent analyses.

Model Evaluation and Assessment

First, the independence model that tests the hypothesis that variables are uncorrelated with one another was rejected, $\chi^2 = 454.75$, $df = 86$, and $p < .000$. Next, the hypothesized model was tested and the chi-square, $\chi^2 = 87.26$, $df = 30$, and $p = .000$, indicates a significant improvement in fit between independence and hypothesized models (see Table 3). However, only marginal support was found for the hypothesized model in terms of the goodness of fit indices. Post hoc model modifications were performed to develop a better fitting and more parsimonious model. The final model provides adequate fit, demonstrates nonsignificant differences between observed and expected covariance matrices, $\chi^2 = 80.66$, $df = 66$, and $p = .106$, and accounts for 59% of the variance. As post hoc model modifications were performed, the correlation coefficient between hypothesized model estimates and estimates from the final model was almost unitary and significant, $r = .97$, $p < .01$. This high correlation indicates that relationships among parameters changed minimally as a result of model modifications. The final model with significant coefficients is shown in Figure 2.

Direct Effects

The accepted model shows that debt is associated positively and significantly with firm size (.24, $SE = .06$), business planning (.14, $SE = .05$), and business objectives (.17,

TABLE 3 Summary of LISREL and Multi-Sample Comparability Results

Model Results	χ^2	<i>df</i>	<i>p</i>	GFI	AGFI	NFI	PFI	RMSR
Hypothesized Family Business Finance Model (Figure 1)	87.26	30	0.000	0.976	0.759	0.808	0.281	0.038
Accepted Family Business Finance Model (Figure 2)	80.66	66	0.106	0.976	0.893	0.822	0.630	0.037
Structural Independence (Null) Model	454.75	86	0.000	0.907	0.677			0.065
Comparability Tests								
Test 1								
Without Imposing Invariance Assumptions	151.37	128	0.078	0.954				0.044
Test 2								
Constraints Imposed on B and Γ	157.80	138	0.097	0.949				0.050

Notes. GFI = goodness of fit; AGFI = adjusted goodness of fit; NFI = normed fit index; PFI = parsimonious fit index; RMSR = root mean square residual.

$SE = .06$). These relationships provide support for hypotheses 1A, 6A, 7A, and 8A and are in the predicted direction. In addition, family control ($.15$, $SE = .06$), which was not hypothesized to have an association with debt, is a significant predictor. These relationships suggest that large family businesses whose owners have a high preference for retaining family control and formal planning practices in place, and whose objective is to create a lifestyle (rather than growth), are more likely to derive their funds from outside entities such as banks, leasing arrangements, and other external funding sources. Age of firm, industry type, and age of CEO are not significant predictors of debt as a source of finance.

Findings reveal that use of family loans is significantly associated with small family businesses ($-.34$, $SE = .06$) and owners who do not have formal planning processes in place ($-.32$, $SE = .06$). In addition, family businesses in the services sector (e.g., retailers and wholesalers) are less likely to use family loans ($-.12$, $SE = .05$) as are those owners who are planning to achieve growth through new product or process development ($-.11$, $SE = .07$). These findings provide support to Harvey and Evan's (1995) contention that given banks' emphasis on the importance of business planning, smaller family businesses that have limited opportunities and training in this area tend to be reliant on internal sources of finance. It is noteworthy that owners who desire retention of family control and age of family firm are not associated with family loans. This could be related to the pecking order notion that a sequencing of funding occurs over the life cycle of a firm. Berger and Udell (1998) found that in the latter stages of the businesses' life cycle, owners choose external debt over insider finance in order to retain ownership and control.

Use of capital and retained profits is likely for family businesses planning to achieve growth through an increase in sales ($.12$, $SE = .04$). In contrast, use of capital and retained profits is less likely for family businesses in the manufacturing sector ($-.08$, $SE = .04$) and lifestyle (cf., growth firms) firms ($-.17$, $SE = .06$). These relationships provide support for hypothesis 8(b), albeit in the opposite direction for hypothesis 2(b). In addition, profitability and retained earnings is related negatively to debt and family loans, providing support for hypothesis 9. However, size of the firm, owners who have a prefer-

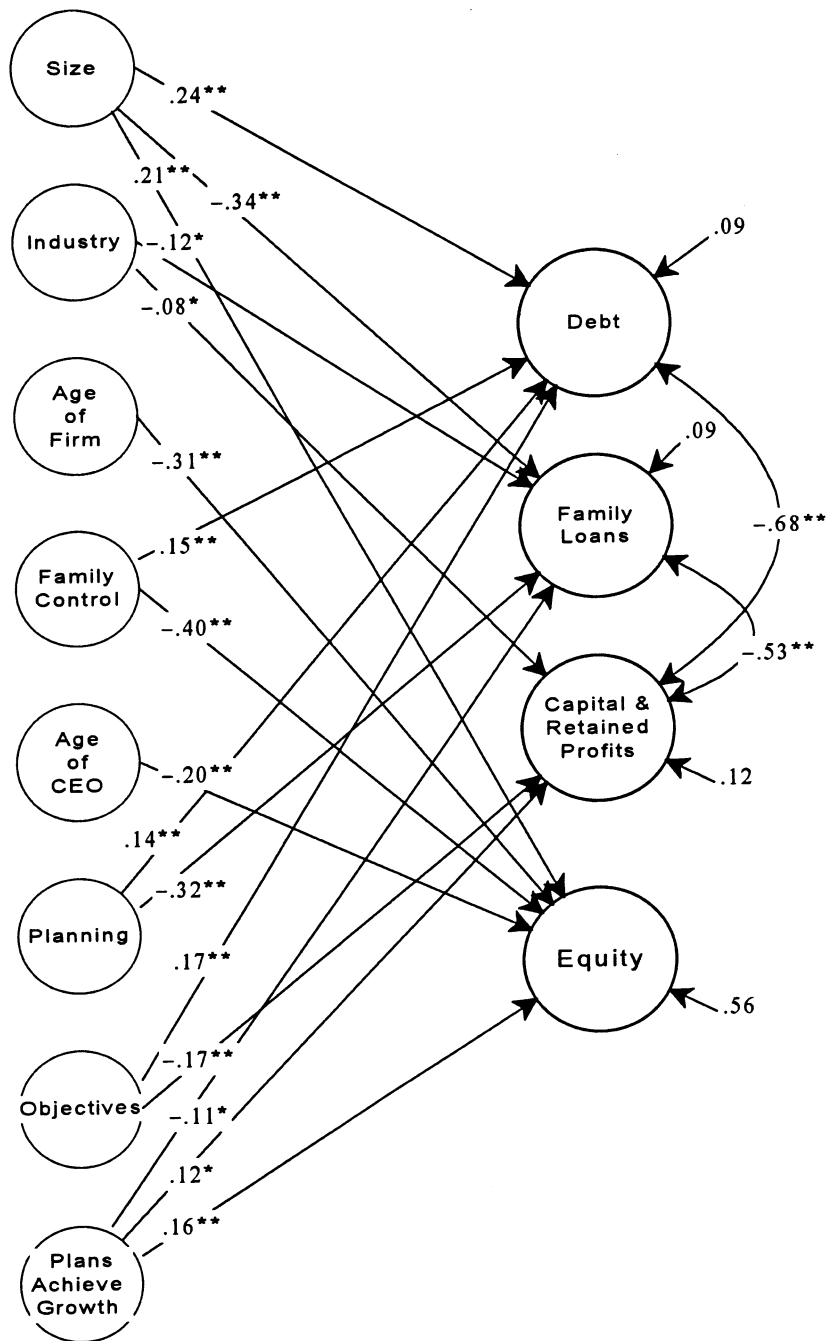


FIGURE 2 Accepted model for family business finance antecedents and outcomes.

ence for retaining family control, age of the owner, and objective was to increase the value of the business were not significant predictors of capital and retained profits.

Equity is a consideration for owners of large businesses (.21, $SE = .07$) and for owners who plan to achieve growth through increasing profit margins (.16, $SE = .06$). However, equity is less likely to be a consideration for older family business owners ($-.20$, $SE = .06$), owners who have a preference for retaining family control ($-.40$, $SE = .07$), and young firms ($-.31$, $SE = .05$). These relationships are in the predicted direction and support hypotheses 1, 4, 5, and 7(b). Industry type and family firms whose objective was to increase the value of their business were not associated with equity.

DISCUSSION

Our study supports findings from a number of investigations (e.g., Bates 1991; Hutchinson 1995; van der Wijst 1989; Waldinger, Aldrich, and Ward 1990; Ward 1987) and demonstrates that size, industry, age of firm, age of CEO, extent of family control, business planning, owners' business objectives, and plans to achieve growth influence family business owners' financing decisions. Moreover, our model has established that family businesses derive their funding from a number of, even multiple, sources; and decisions regarding type of finance are based on a complex array of social, behavioral, and financial factors. For example, our model shows that service industries and firms whose owners' objectives are to create a lifestyle business and who plan to achieve growth through new product or process development are likely to utilize capital and retained profits as a source of business finance.

Our accepted model, which accounts for 59% of the variance, provides empirical support for 11 of our 24 hypotheses and shows that interaction of owner, firm, and family characteristics influence capital structure decision-making processes. Our model supports McMahon and Stanger's (1995) and van der Wijst's (1989) suggestions that business owners' objectives affect their capital structure decision making. A significant positive relationship between size of family business and debt and equity as sources of finance adds weight to Constand, Osteryoung, and Nast's (1991) finding that, when compared with small businesses, large firms tend to use long-term debt. Small family firms' reliance on family loans and debt might be related to owners' interest in retaining control and choosing to set limits on gearing because of risk factors and beliefs that disadvantages of stock exchange listing outweigh its advantages.

Given that most family businesses demonstrate information nontransparency, our results suggest that Modigliani and Miller's (1958) propositions appear to be more relevant to public corporations rather than to family firms (see also Grabowski and Mueller 1972). Pecking Order hypotheses seem to provide useful explanations for family business finance. Pecking Order theory suggests that because of information nontransparency, internally generated funds will be favored by owner-managers. This is particularly the case for owners in small family businesses who tend to be averse to risk and loss of control (Storey 1994). In general, family businesses source capital via external funding in private equity and debt markets rather than via public markets. Unlike larger family firms, smaller family businesses typically have a substantial amount of their funding provided by internally generated funds such as owner capital, other members of the management team, family, and friends.

Notwithstanding this finding, most family businesses are owner-managed and hence agency issues relating to corporate governance and selecting appropriate capital

structures that are driven by ownership and control factors are often irrelevant. However, research (e.g., Berger and Udell 1998; Storey 1994) demonstrates that other factors such as risk aversion or retaining ownership and control of the firm assume prominence in owner-managers' financial decision-making processes. These factors can create their own set of problems.

Poutziouris, Chittenden and Michaelas. (1998) found that a high proportion (50%) of owners of privately-held firms generally avoid sources of finance that weaken links between ownership and control. Of those respondents who would consider using external equity, the three most attractive mechanisms were the Alternative Investment Market, venture capital, and business angels respectively. Moreover, family business owners are averse to pursuing external equity financing because they are reluctant to employ "outside" managers and non-executive directors not drawn from the family group owning the business². Clearly, preferences for a particular form of finance affects a firm's capital structure, growth opportunities, and ultimate long-term survival. The corollary, as demonstrated by negative associations between family control and equity, is that large firm owners who are less concerned with family independence utilize external equity (Constand, Osteryoung, and Nast 1991; Ferri and Jones 1979). Additionally, associations between young businesses that have low levels of family control and capital and retained profits suggest that these owners take advantage of the availability of alternative sources of funds.

In contrast to the findings of Renfrew, Sheehan, and Dunlop (1984), our study demonstrates that small family firms that do not have formalized business plans in writing, tend to derive their funds from family loans. This finding can be explained by Sonnenfeld and Spence's (1989) contention that family business owners are averse to debt and by the possibility of large losses in case of loan failure. Financial institutions also seem to place a heavy reliance on success, wealth, character of the borrower (e.g., integrity and management skills), owners' capacity to repay loans (e.g., cash flow, level of gearing, and business environment), capability of banks to understand and to manage business risk, levels of collateral or security, documentation, and business planning. Berger and Udell (1998) emphasized that banks now want to see a business plan: the blueprint of intentions and expected outcomes. This plan is expected to include an exploration of the business proposal, details of people involved in the business, explanations of how expected outcomes would be reached, descriptions of assumptions on which these outcomes depend, and a SWOT (strengths, weaknesses, opportunities, and threats) analysis. Thus, as argued by Renfrew, Sheehan, and Dunlop (1984) and Riding, Haines, and Thomas (1994), when compared with businesses who have low levels of planning, large firms who demonstrate high levels of wealth and planning capabilities have easier access to funds from financial institutions.

For practitioners, these findings suggest that small family firms are to a large extent averse to using currently regarded nontraditional sources of capital (Mukherjee 1992). Small family businesses tend to rely on family loans for startup and growth. This proclivity is likely to be related to their intentions to retain control and to minimize financial risk. In addition, it is possible that family businesses are not aware of opportunities offered by alternative sources of funding, such as *business angels*, as hitherto these sources traditionally have not directly targeted family firms (Martin and Smith 1997).

In closing, our model provides a framework for understanding capital structure

²We are indebted to one of the blind reviewers for this point.

decision-making processes and has significant theoretical and practical implications. This research tested the propositions of Dreux (1990) and Matthews et al. (1994), which explain relationships between ownership, control, and appropriate levels of capital, along with influences of owners' attitudes, firm characteristics, and environmental constraints. It is noteworthy that our findings are not only in line with Poutziouris, Chittenden, and Michaelas (1998) but have also provided empirical support for the conceptual work of Dreux (1990) and Matthews et al. (1994). This model establishes, quantifiably, the magnitude and direction of relationships between family, business, owner, and industry variables.

Findings of the present study suggest that an important next step is to identify the manner in which contextual variables interrelate and the evolution of specific family variables within broad and global models of firms. Furthermore, investigators should consider assessing differences in antecedent conditions of financing decisions and factors that might be contingent upon specific family variables. In other words, studies should be directed toward developing broad-based theoretical models that demonstrate ways in which relationships between contextual variables affect factors that influence family business owners' decisions to finance their firms.

Our findings raise a number of questions for further research, such as, What are the links between ownership, control, use of external funding, and business performance? This research also suggests a need for cross-national investigations to confirm (or, indeed refute) the model emanating from this investigation.

In conclusion, the development of operational theoretical models not only takes into account complexities associated with how family business owners choose their capital structure but is also conducive to a holistic approach to analysis that allows for the development of integrated, well-founded generalizations regarding dynamic and interactive influences of financing variables. Such an approach uses a process by which theories and hypotheses are tentatively formulated deductively and then tested on data and later reformulated and retested until meaningful outcomes emerge. The present study adopted this approach.

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Appendix 1

Operationalization of Variables

Sources of Family Finance

Sources of finance was measured by four continuous variables: equity (e.g., Renfrew et al., 1984; Riding et al., 1994), debt (Matthews et al. 1994), family loans (e.g., Waldinger et al., 1990), and capital and retained profits (Industry Commission, 1991; National Investment Council, 1995). Respondents were requested to indicate the proportion of funding which was derived from debt, family loans, capital and retained profits, and equity (i.e., Total = 100%).

Size of business. There are a variety of measures of organizational size, of which the three most common operationalizations are natural logarithm of sales, number of employees, and net assets (Scott, 1981). Accordingly, a composite measure based on five continuous variables: number of employees, gross sales, estimated value of business, and number of national and total business locations was employed for this study. This measure was based on 1996 financial ratio data and derived from the PCA.

Business planning. A business plan is a detailed study of an organization's activities, and indicates where an organization has been, where it is now, and where it aims to be in the future (Kotler, 1991; Ward, 1987). Business planning was a composite measure of three variables whether organizations had, in writing, a business plan, a formal strategic long-term plan, and a formal management structure.

Owners' attitude toward family control. Gersick et al. (1997) maintained that ownership is an important dimension of family business and can change over time. These authors noted that many companies are founded and owned by combinations of more than one family generation and can transition from an individual owner-managed business to one in which ownership involves a number of shareholders. *Owners' attitude toward family control* is a composite of three variables (i.e., *Your family might be open to discussion about having non-family shareholders if the conditions were acceptable; Under no circumstances would your family consider making shares available to non-family; Your family business would benefit from external sources of finance which did not involve a full stock exchange listing*). Items were measured on 13-point Likert scales. Thirteen-point scales were used to reduce the likelihood of acquiescent response, to improve reliability levels, and to increase variability in responses so that distributions are less rectangular (see Alwin, 1997; Tabachnick & Fidell, 1996).

Objectives of family business. McMahon and Stanger (1995, p.22) argued that owner-managers' intentions in establishing and running an enterprise can be complex and "unequivocally non-pecuniary". For the present study, owners indicated their most important business objective from the following seven categories: *To accumulate family wealth; to employ family members; to provide family with business careers; to pass onto the next generation; to increase the value of the business; to improve the lifestyle; and other.* Owners also noted how they planned to achieve growth: *By increasing sales; by*

increasing profit margins; through acquisitions; through joint venture(s); through new product/process development; through relocation; and other. Objectives of family business and plans to achieve growth were dummy coded.

Firm age was gauged by the number of years a firm has been in business (i.e., legally registered), while *age of CEO* was measured on an interval scale.

Industry type was assessed using the 10 Australian & New Zealand Standard Industrial Classification (ANZSIC) categories including: Manufacturing, Retail Trade & Wholesale Distribution, Transport & Storage, Finance, Property & Business Services, Recreational Services (e.g., hotels, restaurants, food, and catering), Personal & Other Services, Technology Equipment & Communication Services, Primary Industry (e.g., agriculture, mining), Construction, and Other. However, owing to relatively small numbers in some categories, businesses were reclassified into four main types: manufacturing, services, technology, and other. *Industry* type was dummy coded.